

Vidyut

AI-Native Grid Intelligence for India's Renewable Transition

Concept Document — Pre-Seed / AI Founders Lab

1. One-Line Pitch

Vidyut is a sensor-to-dashboard platform with two AI products: one predicts where and when renewable power will be wasted, the other pinpoints where on the grid energy is being lost to faults or theft. A clamp-on hardware sensor streams live grid data; graph-based AI models reason over that data the way electricity actually flows through the network; a live dashboard shows both what's happening right now and what's about to happen next.

Framing note: this is two AI products, not three separate offerings. The hardware sensor and the dashboard are the input and output layers of each product, not standalone products in their own right. See Section 4.

2. The Problem

India's grid is being asked to absorb renewable power faster than it can flexibly manage it, while simultaneously losing a large share of revenue to technical and commercial losses. Both are large, quantifiable, and currently under-served by AI:

- **Renewable curtailment:** Ember's analysis found India curtailed about 2.1 TWh of renewable electricity in FY 2025-26 to keep coal plants above minimum technical load, representing roughly 1.3% of total renewable generation and about ₹629 crore in foregone revenue. Separately, CREA reported that in Q4 FY26 alone, roughly 31 GW of renewable capacity was curtailed due to transmission constraints and system inflexibility, with peak-hour curtailment reaching 4% of solar and wind generation by April 2026.
- **Technical & commercial (AT&C) losses:** distribution technical losses in India have been estimated around 22.5%, well above the 5-10% international benchmark, and industry estimates put annual revenue loss from power theft nationally in the tens of billions of dollars — even well-run private discoms like Adani Electricity, which cut AT&C losses to about 4.46% in FY 2025-26 through large-scale anti-theft drives, still had to run tens of thousands of manual raids to get there.

Both problems share a root cause: existing tools look at each node (a substation, a meter) in isolation. Neither curtailment nor theft is a single-point phenomenon — they are network phenomena. That's the opening for a graph-native approach.

3. The Solution — Two Products (Not Three)

Both products share the same physical pipeline: a clamp-on hardware sensor streams live data, an AI model interprets it, and a live dashboard delivers the result. The hardware and the dashboard are not separate products — they are how each AI product is delivered. There are two products, defined by the two AI models:

Product 1: Curtailment Prediction

Forecasts, ahead of time, where on the grid a renewable bottleneck will occur — not just how much total generation to expect, but which substations and lines will be constrained and when — so operators can reroute, store, or trade power before curtailment happens instead of reacting after the fact.

Product 2: Fault & Theft Localization

When consumption or voltage anomalies appear across a cluster of meters, this product infers the physical origin point in the network — distinguishing a single point of theft or fault from noisy, correlated signals across many meters — instead of just flagging "something looks off" and leaving field teams to search manually.

4. How Each Product Is Delivered — One Integrated Pipeline

Each product is a single pipeline with three stages: Sense → Predict → Deliver. Read each row across, left to right — it is one flow, not three separate offerings.

4.1 Product 1 Pipeline: Curtailment Prediction

Stage	What Happens	AI or Not
1. Sense (hardware)	A clamp-on sensor at the substation/transformer, plus micro-weather inputs (irradiance, wind speed), streams live grid and weather data continuously. This is live monitoring — the foundation.	No AI — sensor comms only
2. Predict (core IP)	A temporal Graph Neural Network models the grid as a graph — nodes are substations/junctions, edges are transmission lines weighted by capacity — and learns how load and generation propagate across the network over time, forecasting where and when curtailment will occur before it happens.	AI — this is the product
3. Deliver (software)	A live dashboard shows real-time readings alongside forecasted bottlenecks and alerts, plus a track record of predicted-vs-actual outcomes over time.	No AI — deliver only

4.2 Product 2 Pipeline: Fault & Theft Localization

Stage	What Happens	AI or Not
1. Sense (hardware)	A clamp-on CT sensor at transformer/meter clusters streams live voltage and current data continuously — the same live-monitoring foundation as Product 1, reused across the grid's low-voltage network.	No AI — sensor comms only
2. Predict (core IP)	An anomaly detection model (autoencoder / isolation forest) flags unusual consumption or voltage patterns; a graph-based localization model then triangulates the true origin point in the network from multiple correlated anomalous signals, using network topology — not just naming the single worst-looking meter.	AI — this is the product
3. Deliver (software)	A live dashboard flags the location on the network map; field technicians confirm or reject each flag via a mobile app, and that feedback retrain both models over time.	No AI — deliver + feedback capture

Note on edge vs. cloud: predictions can run centrally in the cloud (simpler to build first, requires connectivity) or locally on an edge compute box at the substation (works without a live internet connection, more engineering effort). This is a deployment decision within Stage 2, not a third product.

5. Why This Is Defensible AI, Not Just Infrastructure

Across both pipelines, only the middle stage — Predict — is where genuinely hard, defensible ML work happens. Sense and Deliver exist to feed that stage better data and get its output in front of a human:

- The temporal GNN predicting where and when a bottleneck will occur, using graph structure — not aggregate demand forecasting.
- The graph-based localization model inferring where an anomaly originated from indirect, noisy, multi-point signals — not just flagging the single worst-looking meter.

This is a strength, not a gap: a reviewer will trust "here is the one hard ML problem we solve" far more than a claim that AI is diffusely "everywhere" in the stack. When asked "where's the AI," the answer should point at exactly these two models.

6. Market Opportunity

India added roughly 50 GW of renewable capacity in 2025 alone and is targeting 500 GW of non-fossil capacity by 2030, requiring an estimated ₹30.5 lakh crore in investment through that period. Every gigawatt added increases the curtailment and grid-balancing problem Vidut targets, and India's roughly 60+ state/private distribution companies are simultaneously under regulatory pressure to cut AT&C losses as part of ongoing reform programs (e.g. the national smart meter rollout, targeting 25 crore meters).

- Beachhead customers: state and private DISCOMs, load dispatch centres (SLDCs/RLDCs), and renewable IPPs (independent power producers) managing curtailment-exposed assets.
- Expansion: T&D loss reduction is a budgeted line item for every discom under national reform schemes, making Product 2 a natural add-on sale into the same accounts.

7. Business Model

- **Core revenue: SaaS/B2G licensing per substation-node or per-meter-cluster monitored, sold on annual contracts to discoms and grid operators.**
- **Hardware bundle (optional, lower margin): edge appliance and sensor pods sold or leased alongside the software for sites without existing telemetry.**
- **Value-based pricing angle: price partly against quantifiable savings — avoided curtailment compensation payouts and recovered theft revenue — which gives a concrete, defensible ROI story in sales conversations.**

8. Competitive Landscape

Anomaly detection for meters is close to commodity — several vendors (including smart-meter and AMI platform providers) already offer flagging tools. Vidyut's differentiation is not in detecting that something is wrong, but in graph-based prediction (where/when) and localization (which point in the network) — the two steps that require reasoning over network topology rather than per-node statistics. This is a narrower, more defensible claim than "AI-powered grid analytics" broadly, and it should be stated that way to reviewers and customers alike.

9. Go-to-Market

- Pilot with one mid-size discom or one SLDC on a single feeder/zone, using existing SCADA and smart-meter data — no new hardware required for the first proof of concept.
- Use the pilot's quantified savings (MWh of curtailment avoided, ₹ of theft recovered) as the case study for the next 3–5 sales conversations.
- Layer in the hardware (weather stations, transformer pods) only for sites where existing telemetry is too sparse for the models to perform well.

10. Immediate Next Steps

- Scope the training data pipeline for both core models — what labeled data exists today (SCADA history, past curtailment orders, confirmed theft cases) versus what must be bootstrapped during a pilot.
- Identify 2–3 target discoms or SLDCs for pilot conversations and confirm data-sharing feasibility.
- Pressure-test the "genuinely hard" claim against published literature on graph-based grid forecasting and anomaly localization, so the defensibility story survives technical due diligence.

Note: "Vidyut" (Hindi/Sanskrit for electricity) is the working name direction — confirm domain and trademark availability before locking it in. Market figures cited from Ember, CREA, and public reporting (2025–2026); verify against primary sources before including in investor materials.